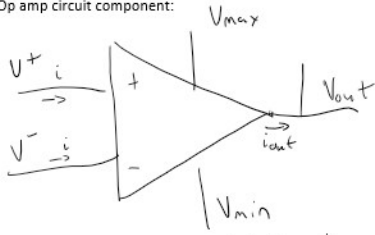


Why use Op-Amps?

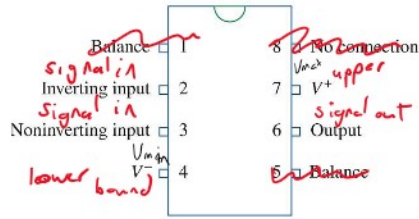
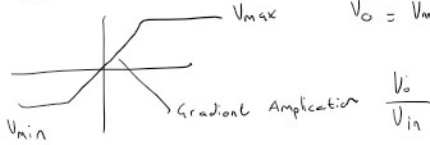
- Amplify
- Isolate

Op amp circuit component:

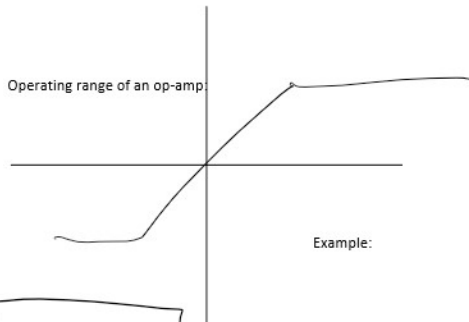
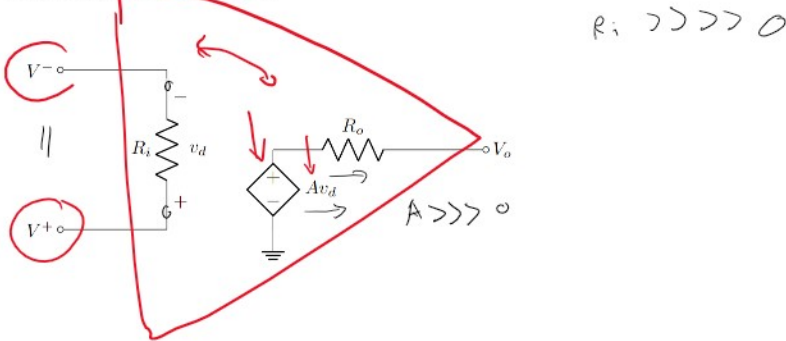


$i = 0$
 $V^- = V^+ = C$
 $i_{out} \neq 0$

if $V_o > V_{max}$
 $V_o = V_{max}$
 if $V_o < V_{min}$
 $V_o = V_{min}$



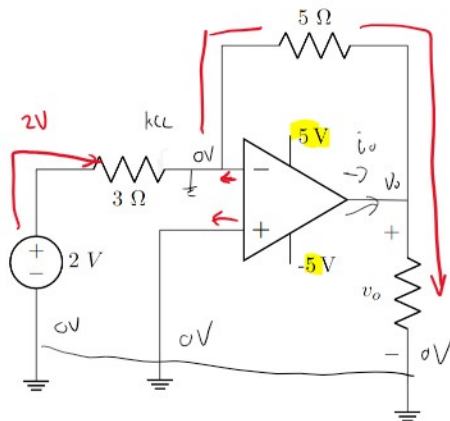
Circuit equivalent for non ideal op amps:



Example:

Summary:

- $V^+ = V^-$
- $i^+ = i^- = 0$
- $V_{max} > V_o > V_{min}$
- i_o is not 0



$$V^+ = V^- = 0V$$

$$V_o \in [-5; 5]$$

$$V_o? \frac{0-2}{3} + \frac{0-V_o}{5} = 0$$

$$V_o = -6V$$